Curriculum Vitae

Kavya Agrawal, PhD Scholar, TERI School of Advanced Studies, New Delhi
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Education

2024 – present	**PhD, Natural & Applied Sciences** TERI School of Advanced Studies, New Delhi. Coursework CGPA 9.0/10.
2021 – 2023	**MA Geography** Dr. B.R. Ambedkar University, Agra. CGPA 7.60.
2018 – 2021	**BA (Honours) Geography** University of Delhi. CGPA 7.86.
2016 – 2018	**Senior Secondary School** Delhi Public School, Aligarh. CGPA 9.80.
2014 – 2016	**Secondary School** St. Francis Inter College, Hathras. CGPA 9.20.

Doctoral research

Thesis	**Development of an Adaptable Machine Learning Based Framework for Multi-Contaminant Prediction**
Supervisor	**Prof. Chander Kumar Singh**
Status	**Full-time. Registered 1 February 2024. Comprehensive examination completed 28 January 2026.**
Coursework	**Research and Publication Ethics (RPR275), A; Research Methodology (RPR271), A; Environmental Statistics (NRE115), A; Digital Image Processing & Information Extraction (NRG172), A**

Research experience

2025 – 26	**Uranium in groundwater across India** *Paper submitted to journal; Best Paper Award, ICSSR EHP NEXUS 2026*
2026 –	**Fluoride and the effect of spatial resolution** *In progress*
2026 –	**Downscaling and upscaling environmental predictors** *In progress*
2024 – 25	**Arsenic in the Middle Gangetic Plain** *Presented at As2024, Bhubaneswar*
2023	**Land surface temperature trends across the Indus River Basin** *Published, Journal of Mountain Science*
2023	**Identification and mapping of permafrost in the Indus Basin**
2023	**Snow cover and changes in Indian basins**

Publications

2023 **Evaluating spatial and elevation-wise daytime/nighttime LST trends across the Indus River Basin**
Mal, S., **Agrawal, K.**, Rani, S. et al. *Journal of Mountain Science* 20, 3154–3172. doi.org/10.1007/s11629-023-8157-8

2023 **Analyzing the Possibilities and Prospects of 24/7 Water Supply in Delhi, India**
Sanu, S. K., Chandrakanta, **Agrawal, K.**, Sharma, V. R., & Bajaj, A. *Journal of Indian Water Works Association*, LV(1). UGC-CARE Group I, ISSN 0970-275X.

Manuscripts

2025–26 **Predicting Uranium Occurrences in Groundwater throughout India**
Research paper
Submitted to journal

2026 **Comparing and Benchmarking Deep Learning Models for Predicting Groundwater Contamination caused by Fluoride with the effects of Spatial Resolution on Prediction**
Research paper
In progress

2026 **A Physically Informed Framework for Multi-Resolution Downscaling and Upscaling of Environmental Predictors (Satellite Data)**
Research paper
In progress

Conference presentations

Apr 2026 **A Machine Learning-Based Framework for Uranium Risk Assessment, Population Exposure and Groundwater Governance in Urban India**
ICSSR EHP NEXUS 2026, Track X: AI and Predictive Analytics for Society and Environment, Gautam Buddha University, 27–28 April 2026. LinkedIn post
First Position, oral presentation, and Best Paper Award

Oct 2024 **Distribution, Speciation and Controls on As Mobilization in the Groundwater in Aquifers of the Middle Gangetic Plain, India**
9th International Congress & Exhibition on Arsenic in the Environment (As2024), KIIT Deemed to be University, Bhubaneswar, 20–24 October 2024. Poster.

Nov 2022 **Problems and Prospects of Delhi's Drinking Water Supply**
IGU Thematic Conference on Sustainability, Future Earth and Humanities, Department of Geography, Central University of Haryana, Mahendragarh, 24–25 November 2022.

Jul 2022 **Educational Relevance of Ramayana in Modern India**
National Seminar on Promoting Human Values and Culture for Peaceful Co-Existence, Amity University Haryana, sponsored by ICPR, 22 July 2022.

May 2022 **Digital Enhancement of Physical Space and Connective Turn in Tourism Industry during and Post-COVID-19 Period**
International Conference on Geography and the Digital: Intersections and Conflations, Department of Geography, Delhi School of Economics, University of Delhi, 23–25 May 2022.

May 2022 **Migrant College Students during Covid Pandemic: Evidence from the Field**
Jamia International Conference on Education 2022, Faculty of Education, Jamia Millia Islamia, New Delhi, 6–8 May 2022.

Apr 2022

Migrant College Students during Covid Pandemic: Evidence from the Field
International Conference on Challenges to Disaster Risk Reduction and Resilient Habitat, Centre for Disaster Management Studies, Shaheed Bhagat Singh College, University of Delhi, 5–6 April 2022.

Teaching

Jun–Dec 2026

Principles of GIS and GNSS (NRG176)
MSc Geoinformatics, TERI School of Advanced Studies. 4 credits, 60 hours (L-T-P 42-2-32).
Course teaching

Jun–Dec 2026

Principles of Remote Sensing (NRG169)
MSc Geoinformatics, TERI School of Advanced Studies. 4 credits, 60 hours (L-T-P 42-2-32).
Course teaching

Jul–Dec 2025

Principles of Remote Sensing (NRG169)
MSc Geoinformatics, TERI School of Advanced Studies. 32 practical learning hours.
Course taught

Projects

Postgraduate, 2021–2023

- Measuring the Problems and Emotions of Urban-Ward Migrants: A Case Study of Construction Workers in Delhi
- Monitoring the Effects of Light Pollution & Evaluating the Effectiveness of Indian Regulations
- Community Awareness, Participation, & CCDRR: Reflections from North-West India
- Experience of Migrant College Students During COVID-19 Pandemic
- Analyzing the Possibilities and Prospects of 24/7 Water Supply in Delhi, India
- Geography of Ramayana
- Digital Enhancement of Physical Space and Connective Turn in the Tourism Industry during and Post-COVID-19 Period

Undergraduate, 2018–2021

- Forest Monitoring & Urban Sprawl Analysis Using GIS in Delhi
- LULC Change Detection for 20 Years in South-West Delhi
- Social Indicators & Status Analysis of Jammu & Kashmir
- Socio-Economic Impact of COVID-19 in Delhi
- Snow Cover & Changes in Indian Basins
- HRVC Analysis of Jim Corbett
- Studying Human Trafficking in Delhi through Remote Sensing
- Understanding Ansal Plaza Mall of South Delhi as a Gendered Landscape: A Temporal Study

Awards and achievements

2026

First Position, oral presentation, and Best Paper Award
ICSSR EHP NEXUS 2026, Gautam Buddha University.

2025

2nd Rank, Map Making Competition
Amity University.

2018

State Child Scientist Award
National Science Congress.

2015 – 2018

Consecutive top 3 ranks, English Olympiads
British Olympiad Foundation.

2016, 2018

Merit Awards

20 cultural and general knowledge awards
From various organisations, including prizes in dance.

Rapporteurship and policy engagement

Mar 2026 — **Rapporteur, National Workshop on Geospatial Technology for Climate Action, Sustainability and Disaster Risk Reduction**
TERI School of Advanced Studies in collaboration with NRSC, ISRO, 16–20 March 2026.

Feb 2026 — **Delegate, World Sustainable Development Summit 2026**
Taj Palace, New Delhi, 25–27 February 2026. Transformations: Vision, Voices, and Values for Sustainable Development and Climate Action.

Mar 2025 — **Rapporteur, Ministerial Session: NDC 3.0 and Multilateralism on the Road to COP30 and COP33**
World Sustainable Development Summit 2025, TERI. India Habitat Centre, New Delhi, 6 March 2025. Session summary

Mar 2025 — **Rapporteur, High-Level Plenary on Achieving Sustainable Development through Integrated Action**
World Sustainable Development Summit 2025, TERI. India Habitat Centre, New Delhi, 7 March 2025. Session summary (with Shweta Gautam)

Positions of responsibility

2026 — **Facilitator and Judge, Programme LEAD 2026**
TERI SAS. Sustainable City Design Challenge for students of classes 9–12.

Nov 2025 — **Judge and Panel Member, TERI SAS Sustainability Hackathon 2025**
Expert evaluation panel for school and undergraduate sustainability proposals, 18–19 November 2025.

2025 — **Judge, KnoxCode 2025, Geospatial Innovation Challenge**
Organised by GeoknoX, TERI SAS, as part of Aahwaan Fest 2025.

2018 – 2021 — **Editor-in-Chief, Geography Association**
University of Delhi.

2018 – 2020 — **Magazine Coordinator, *Vani***
Shaheed Bhagat Singh College, University of Delhi.

2019 — **English Editor, *Vani***
Shaheed Bhagat Singh College, University of Delhi.

Workshops, seminars and summits attended

Feb 2026 — **Water as a Commons workshop, Water Governance Working Group**
With the Ostrom Workshop, Indiana University Bloomington and IU India Gateway, 26–27 February 2026; included a Yamuna River field visit with INTACH Delhi.

Feb 2026 — **Ambition to Action: Climathon, Delhi Climate Innovation Week**
Climate Career & Sustainability Expo, Innovate X Delhi, 21 February 2026.

Feb 2026 — **Guest lecture: Soil Carbon Sequestration as a Climate Mitigation Mechanism**
Ms. Violeta Martinkė, Project Development Lead, InSoil, 13 February 2026.

Oct 2025 — **Act4Earth Dialogue: Strengthening Multilateralism on the Road to COP30 and Beyond**
Virtual, 17 October 2025.

Oct 2025	**Expert talk: DHI Urban Water Solutions** Mr. Morten Just Kjølby, Product Portfolio Manager for MIKE+, DHI Denmark, TERI SAS, 7 October 2025.
Sep 2025	**Science Summit 2025: Marine Resources for Sustainable Development** 16 September 2025.
Aug 2025	**QGreenland Educator Workshop** Virtual, 5–7 August 2025. Developing educational activities with QGreenland data.
Jul 2025	**Roots to Resilience Seminar** TERI SAS with UNEP, UNESCO and Akashvani, 25 July 2025.
Mar 2025	**World Sustainable Development Summit 2025** New Delhi, 5–7 March 2025. Partnerships for Accelerating Sustainable Development and Climate Solutions.
Oct 2024	**Lecture by Professor John A. Cherry on groundwater and sustainable development** 2020 Stockholm Water Prize laureate, 24 October 2024. Prof. Cherry also spent four hours at the Analytical and Geochemistry Laboratory, TERI SAS, discussing the lab's current and future work.
	Climate Skills: Seeds for Transition TERI SAS and TERI with British Council and HSBC Bank.

Training, capacity building and schools

Mar 2026	**National Workshop on Geospatial Technology for Climate Action, Sustainability and Disaster Risk Reduction** TERI SAS in collaboration with NRSC, ISRO, 16–20 March 2026. Participant, volunteer and rapporteur.
Dec 2024	**Capacity Building Workshop on UAV/Drone Applications in Agriculture and Natural Resource Management** TERI SAS with IIT BHU, IIT Roorkee, Namami Gange, Hokkaido University, GLP Earth, GLP Japan, Abhipsa Foundation and Kambil, 10–12 December 2024.
Nov – Dec 2024	**21-day Winter School Capacity Building Program in Geospatial Science and Technology** Amity University, Noida, supported by the National Geospatial Program, DST, Government of India, 18 November – 7 December 2024. Included a visit to Esri India.
Mar 2024	**Harnessing Geospatial Technology for Forest Fire Disaster Management: Mapping, Monitoring and Mitigation** One-week capacity building programme, Amity University, Noida, 11–15 March 2024.

Skills

GIS and remote sensing

Google Earth Engine (JavaScript), ArcGIS / ArcGIS Pro, QGIS, ERDAS Imagine, Google Earth; image enhancement and feature extraction, orthorectification and mosaicking, DEM processing, satellite image acquisition and processing.

Programming and machine learning

Python, pandas; machine learning, deep learning and AutoML (AutoGluon); Random Forest, LightGBM; explainable AI (SHAP, LIME).

Spatial analysis and statistics

Spatial data analysis and visualisation; time-series modelling and trend analysis; Mann–Kendall test, Theil–Sen slope; Moran's I, semivariograms.

Earth observation and environmental data

Landsat 8/9, Sentinel, MODIS, VIIRS, FIRMS, CHIRPS, IMD, ERA5-Land, TerraClimate, GLDAS, GRACE, SRTM, MERIT Hydro, JRC Global Surface Water,

OpenLandMap, GPW, GHM, GHSL, SMOD, LCZ; CGWB hydrochemical observations.

Parameters

MAAT, BMAT, PISR, land surface temperature, NDVI/NDBI, night-time light, snow, precipitation, slope, aerosols and solar radiation, hydrochemistry (pH, EC, major ions, nitrate, fluoride, uranium).

Data quality

Provenance and lineage, duplicate and sentinel-value handling, coordinate validation, unit checks, resolution harmonisation, independent validation, masked-pixel handling.

Research and field

Field and survey-based research, socio-environmental data analysis, drinking-water quality assessment, disaster risk reduction, climate–water–environment interactions, sustainability and climate policy.

Professional

Academic writing and conference presentations, scientific communication and reporting, editorial and content development, rapporteurship and policy documentation, academic coordination, interdisciplinary collaboration.

Volunteering and co-curricular

2016 – present	**Volunteer, SWAPO NGO**
2019 – 2021	**National Service Scheme (NSS)**
2018 – 2019	**Coordination of Volunteers, Vasundhara**
2018 – 2023	**Personal blogging and content writing**

Interests

Collecting old coins and stamps, reading historic stories and accounts, photography and videography, travelling, badminton and dance.

Contact

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